DATABASE SCHEMA REFERENCE

Generated on: 2025-10-14
Database: psmenow_natcon
Host: 20.24.15.42

Overview

This document provides a comprehensive reference for the PSME National Convention database schema. The database supports a food registration and check-in management system with external API integration capabilities.

TABLES

1. `food` Table

Purpose: Stores food registration records for the national convention

Column	Type	Null	Key	Default	Extra	Description
`id`	int(11)	NO	PRI	NULL	auto_increment	Primary key, unique identifier
`email_address`	varchar(255)	NO	UNI	NULL		Registrant's email address (unique)
`first_name`	varchar(100)	NO		NULL		Registrant's first name
`middle_name`	varchar(100)	YES		NULL		Registrant's middle name (optional)
`last_name`	varchar(100)	NO		NULL		Registrant's last name
`suffix`	varchar(20)	YES		NULL		Name suffix (Jr., Sr., III, etc.)
`mapping`	varchar(255)	YES		NULL		Email mapping for external API correlation

Indexes:

- PRIMARY KEY (`id`)
- UNIQUE KEY `email_address` (`email_address`)

Relationships: None (standalone table)

Business Rules:

- Email address must be unique across all registrations
- First name, last name, and email address are required
- Mapping field is used for correlating with external PSME API data
- When mapping is NULL or empty, the record is considered "unmapped"

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2. `checkins` Table

Purpose: Stores check-in records and kit release information

Column	Type	Null	Key	Default	Extra	Description
`id`	int(11)	NO	PRI	NULL	auto_increment	Primary key, unique identifier

Column	Type	Null	Key	Default	Extra	Description
member_id	varchar(50)	YES		NULL		Member identification number
first_name	varchar(100)	NO		NULL		Member's first name
last_name	varchar(100)	NO		NULL		Member's last name
email_address	varchar(255)	YES		NULL		Member's email address
checkin_time	datetime	YES		NULL		Timestamp when member checked in
kit_release_time	datetime	YES		NULL		Timestamp when kit was released
qr_code	varchar(255)	YES		NULL		QR code for member identification
status	enum('pending','checked_in','kit_released')	YES		'pending'		Current status of the member

Indexes:

- PRIMARY KEY (id)
- INDEX member_id (member_id)
- INDEX email_address (email_address)

Business Rules:

- checkin_time is set when member checks in
- kit_release_time is set when kit is released to member
- Status progresses: pending → checked_in → kit_released
- QR codes are used for quick identification during check-in process

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3. users Table

Purpose: Stores system user accounts and authentication data

Column	Type	Null	Key	Default	Extra	Description
id	int(11)	NO	PRI	NULL	auto_increment	Primary key, unique identifier
username	varchar(50)	NO	UNI	NULL		Unique username for login
password_hash	varchar(255)	NO		NULL		Hashed password
email	varchar(255)	YES		NULL		User's email address

Column	Type	Null	Key	Default	Extra	Description
role	enum('admin','user')	YES		'user'		User role/permission level
access_token	varchar(500)	YES		NULL		Session/API access token
created_at	timestamp	YES		CURRENT_TIMESTAMP		Account creation timestamp
last_login	timestamp	YES		NULL		Last successful login timestamp

Indexes:

- PRIMARY KEY (`id`)
- UNIQUE KEY `username` (`username`)

Business Rules:

- Username must be unique
- Passwords are stored as hashed values (never plain text)
- Access tokens are used for API authentication and session management
- Admin role has full system access, user role has limited access

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API INTEGRATION

External PSME API

Endpoint: https://psme-api.smartpay.ph/public/index.php/api/postSearch
Method: POST
Authentication: Bearer token (stored in users.access_token)
Purpose: Look up member information for food registration mapping

Parameters:

- `search`: Email address to search for
- `event_id`: Event identifier (currently using 16)

Response Structure:

```
{
    "resultKey": 1,
    "resultValue": [
        {
            "firstname": "John",
            "lastname": "Doe",
            "email": "john.doe@example.com"
        }
    ],
    "errorCode": 0
}
```

ENDPOINTS REFERENCE

Food Registration Management

- **GET** `fetchFoodData` - DataTables server-side processing
- **POST** `fetchAllFoodData` - Export all food data
- **GET** `foodData` - Dashboard statistics (total/unmapped counts)
- **POST** `addFoodRegistration` - Add new food registration
- **GET** `getFoodRegistration?id=X` - Get single registration for editing
- **POST** `updateFoodRegistration` - Update existing registration
- **POST** `deleteFoodRegistration` - Delete registration
- **GET** `getUnmappedFoodData` - Get records without mapping
- **POST** `updateFoodMapping` - Update mapping field for registration

Check-in Management

- **GET** `data` - Dashboard statistics (checkins/kit releases)
- **POST** `fetchCheckinsData` - DataTables processing for checkins
- **POST** `insertCheckinData` - Add new check-in record
- **POST** `deleteCheckin` - Remove check-in record

Authentication & Session

- **POST** `loginScript` - User authentication
- **GET** `renewToken` - Refresh API access token
- **GET** `logout` - End user session

DASHBOARD METRICS

Food Registration Dashboard

1. **Total Food Registration**: `SELECT COUNT(*) FROM food`
2. **Total Unmapped**: `SELECT COUNT(*) FROM food WHERE mapping IS NULL OR mapping = ''`

Check-in Dashboard

1. **Total Checked In**: `SELECT COUNT(*) FROM checkins WHERE checkin_time IS NOT NULL`
2. **Total Released Kits**: `SELECT COUNT(*) FROM checkins WHERE kit_release_time IS NOT NULL`

FILE STRUCTURE

Core Files

- `dbConnection.php` - Database connection configuration
- `functions.php` - Common utility functions
- `sidenav.php` - Navigation sidebar component
- `topbar.php` - Header/topbar component

Food Registration System

- `list_food.php` - Main food registration interface
- `addFoodRegistration.php` - Add new registration endpoint
- `updateFoodRegistration.php` - Update registration endpoint
- `deleteFoodRegistration.php` - Delete registration endpoint
- `getFoodRegistration.php` - Get single registration endpoint
- `fetchFoodData.php` - DataTables data endpoint
- `fetchAllFoodData.php` - Export data endpoint
- `foodData.php` - Dashboard statistics endpoint
- `getUnmappedFoodData.php` - Unmapped records endpoint
- `updateFoodMapping.php` - Mapping update endpoint

Check-in System

- `checkin.php` - Main check-in interface
- `list_checkins.php` - Check-in records management
- `insertCheckinData.php` - Add check-in endpoint
- `deleteCheckin.php` - Delete check-in endpoint
- `fetchCheckinsData.php` - DataTables data endpoint
- `data.php` - Dashboard statistics endpoint

Authentication

- `login.php` - Login interface
- `loginScript.php` - Authentication processing
- `logout.php` - Logout processing
- `renewToken.php` - Token renewal endpoint

BUSINESS LOGIC

Food Registration Workflow

1. **Registration Entry**: Add new food registrations via web interface
2. **Mapping Process**: Correlate local registrations with external PSME API data
3. **Bulk Operations**: Map multiple registrations automatically using API lookup
4. **Data Management**: Edit/delete registrations as needed
5. **Reporting**: Export data for external processing

Check-in Workflow

1. **Member Arrival**: Scan QR code or manual entry for check-in
2. **Status Update**: Update member status to "checked_in"
3. **Kit Release**: Mark kit as released when distributed
4. **Status Tracking**: Monitor check-in and kit release statistics

Authentication Flow

1. **Login**: Username/password authentication
2. **Session Management**: Access token-based sessions
3. **Token Renewal**: Automatic token refresh for external API access
4. **Permission Control**: Role-based access (admin/user)

TECHNICAL NOTES

Server Compatibility

- **Production**: Apache with mod_rewrite (.htaccess routing)
- **Development**: Python HTTP server (limited PHP GET parameter support)
- **Database**: MySQL/MariaDB with utf8mb4 charset
- **PHP Version**: Compatible with PHP 7.4+

Security Features

- **Authentication**: Session-based with access tokens
- **SQL Injection Protection**: Prepared statements throughout
- **Input Validation**: Server-side validation for all inputs
- **CSRF Protection**: Session validation on all POST operations
- **Email Uniqueness**: Enforced at database and application level

Performance Considerations

- **DataTables**: Server-side processing for large datasets
- **Indexing**: Proper indexes on frequently queried columns
- **Caching**: Disabled for AJAX requests to ensure fresh data
- **Pagination**: 25 records per page default with customizable options
- **API Rate Limiting**: 200ms delay between external API calls

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This reference document should be updated whenever database schema changes are made.